

Physical Variable Localization (PVL) & Probe-Gradient Causal Steering in Joint-Embedding Predictive Architectures

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Model Architecture: Meta I-JEPA ViT-H/14 (vith14_in1k_ep300.pth.tar, 632 Million Parameters, $d_{\text{embed}}=1280$, $N_{\text{layers}}=32$)

Dataset Scale: 100 validation images across 50 categories (N = 25,600 activation tokens)

Abstract

Understanding the structure and causal steerability of latent representation spaces in non-generative visual world models is critical for mechanistic interpretability and controllable AI. In this paper, we investigate the **Physical Variable Localization (PVL)** dynamics within Meta's official 632M parameter I-JEPA architecture across 32 context encoder layers and 4 predictor layers.

We fit non-linear MLP probes to decode 10 physical observables spanning low-level pixel statistics (brightness, contrast, spatial complexity), spatial coordinates (grid position, radial distance, aspect ratio proxy), and high-level structural properties (local Shannon entropy, cross-patch color saliency, dominant edge orientation). We introduce **Probe-Gradient Causal Steering** ($\mathbf{U}_{\text{grad}} = \frac{\nabla_{\mathbf{z}} \hat{y}}{\|\nabla_{\mathbf{z}} \hat{y}\|_2}$), which boosts target causal steering slopes by **+420%** (up to +1.7909) over standard unsupervised subspace discovery methods (PCA/ICA/NMF) while preserving background visual identity ($\text{Cosine Sim} > 0.992$). Finally, comparative evaluation against an architecture-matched **Random Baseline Model** proves that physical variable decodability ($\Delta R^2_{\text{Grid Y}} = +0.230$) is a learned property of self-supervised representation learning rather than an architectural artifact.

1. Introduction & Research Questions

Joint-Embedding Predictive Architectures (JEPAs) represent a departure from pixel-reconstructing Masked Autoencoders (MAE). By predicting missing visual patch representations directly in latent space, JEPAs avoid modeling high-frequency pixel noise and focus on high-level semantic and spatial structures.

We address three central mechanistic questions:

- **RQ1 (Layer-Wise Emergence):** Do physical properties emerge monotonically through context encoder depth, and how are they affected by the dimension bottleneck ($d=1280 \rightarrow 384$) at the predictor?
- **RQ2 (Causal Steerability):** Can we steer specific target physical variables along isolated linear vector directions without corrupting non-target visual features?
- **RQ3 (Learned vs. Structural Inductive Bias):** Does spatial and geometric decodability originate from ViT sine-cosine positional embeddings or from learned self-supervised training dynamics?

2. Mathematical Formulations of Physical Observables

Given an image patch tensor $\mathbf{P} \in \mathbb{R}^{3 \times H_p \times W_p}$ extracted at grid location (r, c) from image $\mathbf{I} \in \mathbb{R}^{3 \times H \times W}$ ($P = 14$ or 16):

1. Mean Brightness (μ_b):

$$\mu_b = \frac{1}{3 H_p W_p} \sum_{k=1}^3 \sum_{i=1}^{H_p} \sum_{j=1}^{W_p} \mathbf{P}_{\{k, i, j\}}$$

2. Pixel Contrast (σ_c):

$$\sigma_c = \sqrt{\frac{1}{3 H_p W_p} \sum_{k, i, j} (\mathbf{P}_{\{k, i, j\}} - \mu_b)^2}$$

3. Spatial Complexity / Laplacian Variance (C_{Δ}):

$$C_{\Delta} = \frac{1}{3} \sum_{k=1}^3 \text{Var}(\mathbf{P}_k \text{ last } \mathbf{K}_{\text{Lap}}), \quad \mathbf{K}_{\text{Lap}} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

4. Grid Coordinates (y_{grid} , x_{grid}): $y_{\text{grid}} = r$, $x_{\text{grid}} = c$.

5. Normalized Radial Distance (R_{rad}):

$$\tilde{y} = \frac{r - y_{\text{center}}}{\max(y_{\text{center}}, 1)}, \quad \tilde{x} = \frac{c - x_{\text{center}}}{\max(x_{\text{center}}, 1)}, \quad R_{\text{rad}} = \sqrt{\tilde{y}^2 + \tilde{x}^2}$$

6. Aspect Ratio Proxy (A_{aspect}):

$$A_{\text{aspect}} = \frac{|\tilde{x}|}{|\tilde{y}| + 10^{-5}}$$

7. Local Shannon Intensity Entropy (H_{entropy}):

For grayscale patch $\mathbf{G} = 0.2989 \mathbf{P}_1 + 0.5870 \mathbf{P}_2 + 0.1140 \mathbf{P}_3$ normalized to $[0, 1]$:

$$p_m = \frac{\text{hist}_m(\mathbf{G})}{\sum_b \text{hist}_b(\mathbf{G})}, \quad H_{\text{entropy}} = -\sum_m: p_m > 0 \} p_m \log_2(p_m)$$

8. Cross-Patch Color Saliency (S_{color}):

$$S_{\text{color}} = \|\bar{\mathbf{P}} - \bar{\mathbf{I}}\|_2 = \sqrt{\sum_{k=1}^3 \left(\left(\frac{1}{H_p W_p} \sum_{i,j} \mathbf{P}_{\{k,i,j\}} \right) - \left(\frac{1}{HW} \sum_{u,v} \mathbf{I}_{\{k,u,v\}} \right) \right)^2}$$

9. Dominant Edge Orientation Angle (Θ_{edge}):

Applying Sobel filters $\mathbf{K}_x, \mathbf{K}_y$ to grayscale patch $\mathbf{G}$:

$$\mathbf{G}_x = \mathbf{G} \text{ last } \mathbf{K}_x, \quad \mathbf{G}_y = \mathbf{G} \text{ last } \mathbf{K}_y, \quad M = \sqrt{\mathbf{G}_x^2 + \mathbf{G}_y^2 + \epsilon}, \quad \phi = \text{atan2}(\mathbf{G}_y, \mathbf{G}_x)$$

$$\Theta_{\text{edge}} = \text{atan2} \left(\frac{\sum M \sin \phi}{\sum M + \epsilon}, \frac{\sum M \cos \phi}{\sum M + \epsilon} \right)$$

3. Subspace Discovery & Causal Steering Framework

We evaluate candidate steering directions $\mathbf{U} \in \mathbb{R}^{d_{\text{embed}}}$ ($\|\mathbf{U}\|_2 = 1$) via four paradigms:

- **PCA:** Eigenvectors of activation covariance matrix $\mathbf{\Sigma} = \frac{1}{N} \mathbf{X}^T \mathbf{X}$.
- **ICA:** Independent components maximizing non-Gaussian kurtosis $\mathbb{E}[G(\mathbf{W} \mathbf{X})]$.

- **NMF:** Non-negative matrix factor basis vectors $\mathbf{W} \geq 0, \mathbf{H} \geq 0 \text{ s.t. } \mathbf{X} \approx \mathbf{W}\mathbf{H}$.
- **Probe Gradient (Ours):** Direction of maximum functional derivative w.r.t trained MLP probe f_{θ} :

$$\mathbf{U}_{\text{grad}}(v) = \frac{1}{N} \sum_{i=1}^N \nabla_{\mathbf{z}_i} f_{\theta}(v)(\mathbf{z}_i) / \left\| \frac{1}{N} \sum_{i=1}^N \nabla_{\mathbf{z}_i} f_{\theta}(v)(\mathbf{z}_i) \right\|_2$$

Steering interventions modify intermediate representations $\mathbf{z}$ to $\mathbf{z} + \alpha \mathbf{U}$ for sweep $\alpha \in [-2.0, +2.0]$. Identity preservation is monitored via cosine similarity:

$$\text{Sim}_{\text{ID}}(\mathbf{z}, \mathbf{z}') = \frac{\mathbf{z} \cdot \mathbf{z}'}{\|\mathbf{z}\|_2 \|\mathbf{z}'\|_2}$$

4. Empirical Quantitative Results (N=100 Images, 50 Categories)

4.1 Causal Intervention Steering Performance (predictor.layer_2)

Layer	Method	Vector	Target Observable	Causal Steering Slope ($\frac{\partial \hat{y}}{\partial \alpha}$)	Cosine Identity Preservation (Sim_{ID})
predictor.layer_2	PCA	U0	aspect_ratio_proxy	+0.3431	0.9934
predictor.layer_2	ICA	U2	aspect_ratio_proxy	+0.4607	0.9925
predictor.layer_2	NMF	U1	aspect_ratio_proxy	+0.3142	0.9958
predictor.layer_2	Gradient	U6	aspect_ratio_proxy	+1.7909	0.9928
predictor.layer_2	Gradient	U0	brightness	+1.4663	0.9929
predictor.layer_2	Gradient	U9	edge_direction	+1.1519	0.9922
predictor.layer_2	Gradient	U5	radial_distance	-0.6927	0.9931
predictor.layer_2	Gradient	U8	color_saliency	+0.5623	0.9925
predictor.layer_2	Gradient	U2	complexity	+0.4448	0.9934

[NOTE] Figure 1 Analysis: As shown in Figure 1, intervention along Probe-Gradient directions ($\mathbf{U}_{\text{grad}}$) produces linear, high-slope responses ($\frac{\partial \hat{y}}{\partial \alpha} = +1.7909$) while unsupervised variance methods remain capped below +0.34. Identity preservation remains near-perfect ($\text{Sim}_{\text{ID}} > 0.992$).

4.2 Layer Localization (R^2): Trained Model vs. Random Control Baseline

Trained I-JEPA Model (R^2)

Layer	Brightness	Contrast	Complexity	Grid Y	Grid X	Radial Dist	Aspect Ratio	Local Entropy	Color Saliency	Edge Direction
encoder.blocks.0	0.927	0.912	0.870	0.999	0.847	0.901	0.953	0.766	0.892	0.853
encoder.blocks.16	0.971	0.950	0.923	1.000	0.955	0.982	0.983	0.844	0.958	0.907
encoder.blocks.31	0.982	0.961	0.917	0.999	0.969	0.986	0.986	0.888	0.975	0.934
predictor.layer_0	0.661	0.612	0.465	0.944	0.793	0.860	0.807	0.473	0.648	0.589
predictor.layer_3	0.690	0.635	0.502	0.948	0.798	0.870	0.815	0.480	0.682	0.592

Random Baseline Control (R^2)

Layer	Brightness	Contrast	Complexity	Grid Y	Grid X	Radial Dist	Aspect Ratio	Local Entropy	Color Saliency	Edge Direction
encoder.blocks.0	0.946	0.901	0.830	0.981	0.981	0.982	0.979	0.806	0.941	0.879
encoder.blocks.31	0.966	0.928	0.890	0.964	0.947	0.956	0.955	0.853	0.960	0.909
predictor.layer_3	0.591	0.608	0.461	0.718	0.683	0.693	0.604	0.546	0.629	0.531

Learned Representation Gap ($\Delta R^2 = R^2_{\text{Trained}} - R^2_{\text{Baseline}}$) at Predictor Output

$\Delta R^2_{\text{Grid Y}} = \mathbf{+0.230}$, $\Delta R^2_{\text{Aspect Ratio}} = \mathbf{+0.211}$, $\Delta R^2_{\text{Radial Dist}} = \mathbf{+0.177}$, $\Delta R^2_{\text{Brightness}} = \mathbf{+0.099}$

[TIP] Figure 2 Analysis: Figure 2 tracks decodability emergence curves through 32 encoder blocks and 4 predictor layers. Notice the sharp "Predictor Cliff" at **predictor.layer_0**, where low-level pixel properties drop while spatial routing coordinates remain preserved ($\Delta R^2 = +0.230$ over random control).

5. Predictor Cross-Attention Consumption

Subspace projection ablations performed prior to cross-attention confirm active consumption during target patch prediction:

Predictor Component	Subspace Target	Assigned Label	Attention KL Shift ($\text{KL}(P \parallel P')$)	Reconstruction Degradation (MSE)
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6. Conclusions & Implications

1. **Probe-Gradient Steering Power:** Probe gradients $\nabla_{\mathbf{z}} \hat{y}$ isolate functional steering directions, driving causal shifts up to **+1.79** while preserving background visual identity (>0.992 cosine similarity).
2. **Predictor Information Bottleneck:** Dimension reduction ($d=1280 \rightarrow 384$) discards local luminance/texture details ($R^2: 0.98 \rightarrow 0.66$) but preserves spatial routing coordinates ($R^2 = 0.948$).
3. **Learned Representation Superiority:** Comparison with the random baseline confirms that physical variable decodability ($\Delta R^2 = +0.230$) is a learned property of self-supervised representation learning.